Pandora (reconstruction) experience with the IDEA detector

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CERN

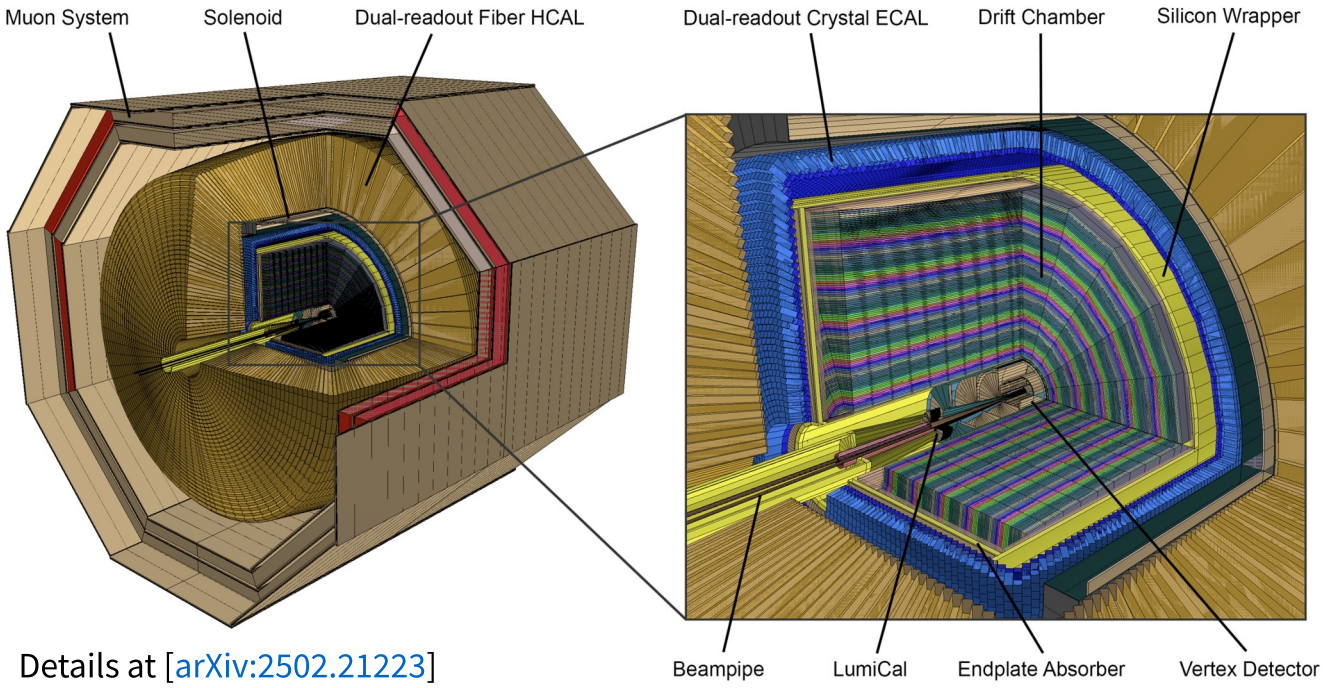




Introduction

The IDEA detector

- Silicon vertex detector
- Ultra-light drift chamber
  - Excellent PID with the cluster counting technique (dN/dx)
- (For option 2) crystal dual-readout
  - Excellent EM resolution
  - Crystal dual-readout allowing application of the dual-readout principle for hadronic showers
- Thin & high-temperature solenoid
- Fiber dual-readout calorimeter
- µRWELL muon chamber



Details at [arXiv:2502.21223]

IDEA option 2

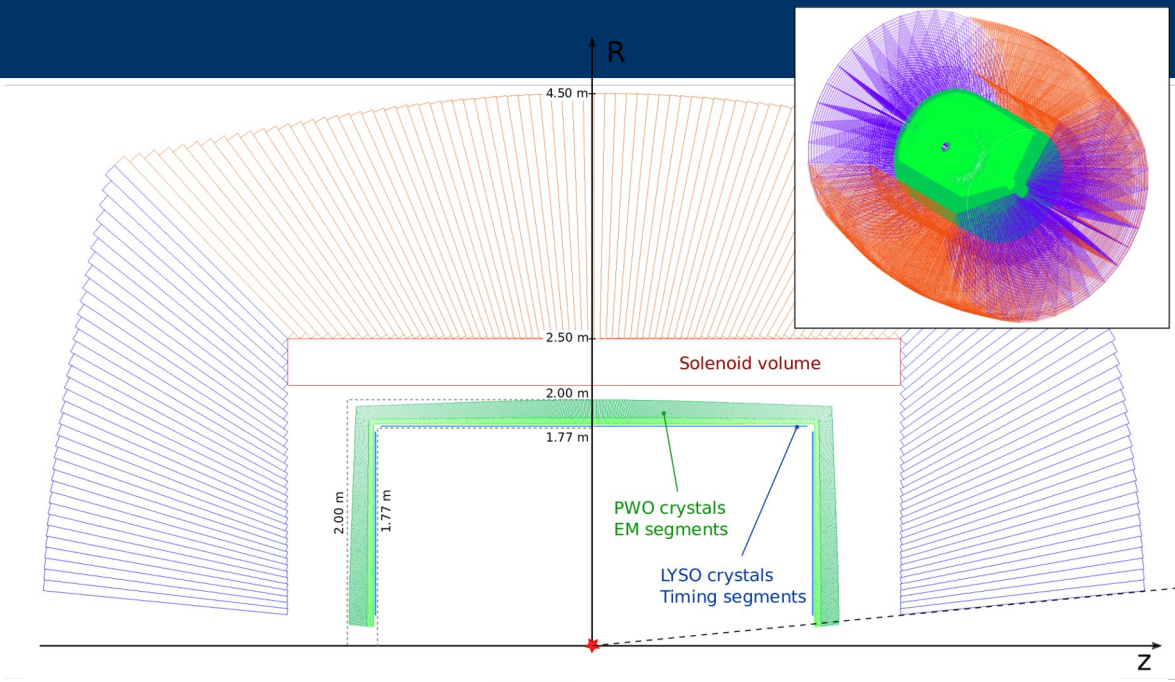
Crystal dual-readout calorimeter

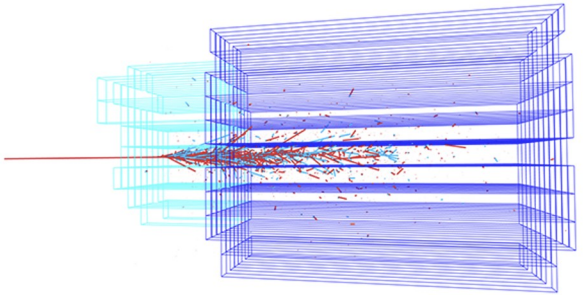
- The main difference between the two options is the crystal dual-readout ECAL
  - Provides an excellent EM energy resolution
  - Two types of sensors to enable the dual-readout principle by combining with the fiber dual-readout HCAL

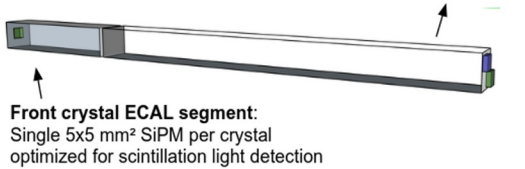
    (hadrons also deposit energies in ECAL)

- Implemented DD4hep geometry recently
- Further optimization needs to be done to reduce the computation resource requirement

- (standalone) particle flow studies based on the option 2 are available at [arXiv:2202.01474]







IDEA option 1

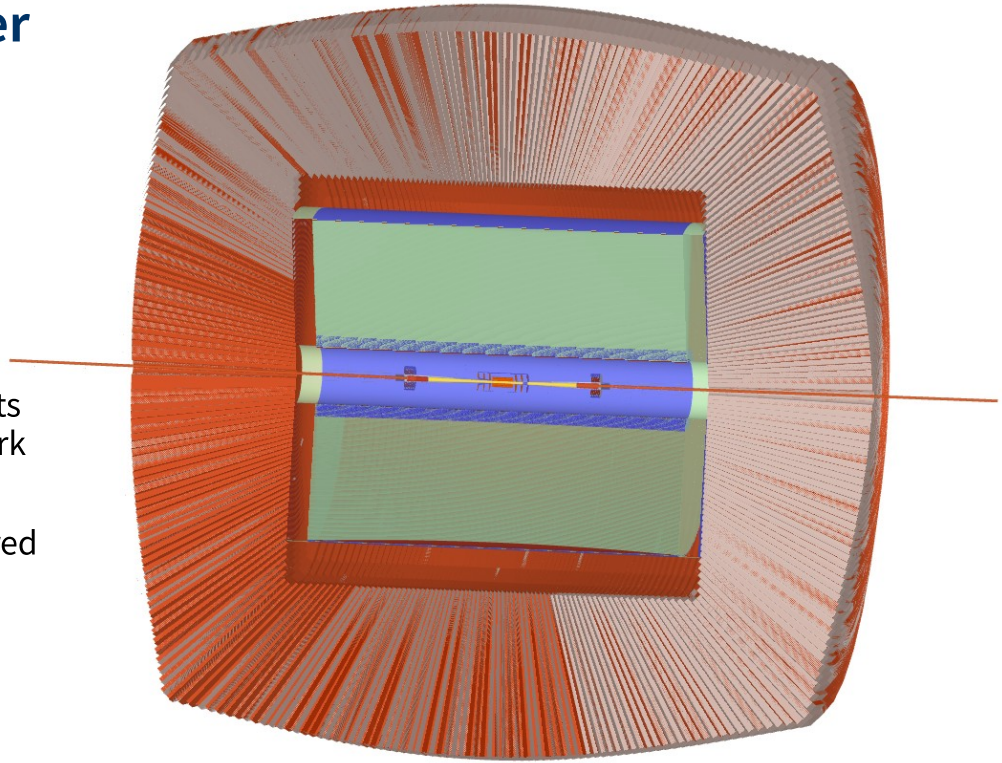
Monolithic fiber dual-readout calorimeter

- The option 1 consists of the "monolithic" dual-readout calorimeter that measures both EM and hadronic showers simultaneously

  (closer to the initial "the 4th concept detector")

- Integration with Key4hep and the (computing) performance optimization had been done
- Possible to run a reasonable number of full simulation events that allow reconstruction studies using the central framework

- Note: following studies do not target developing a PFA tailored to the specific option
- rather to establish a (crude) overall structure and building blocks that can be shared for both options



Tracking

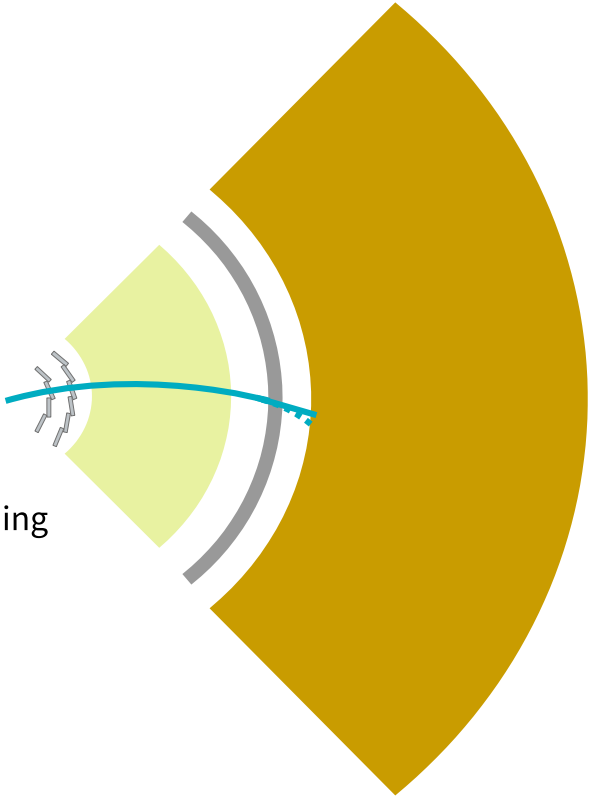
Current track reconstruction scheme

- Plan to implement Kalman filter (KF) based track reconstruction using GenFit
- Integration of GenFit with the IDEA drift chamber is still ongoing
  → Utilizing "truth" track as a workaround for particle reconstruction studies

- Track parameters taken from the generated particle (GenPtc)
- Incorporate SimTrackerHits associated to the GenPtc to the truth track
- Extrapolate to the calo surface based on the simple helix hypothesis

- Caveat: IDEA option 1 has a solenoid between the calo and the drift chamber
  → Need more realistic trajectory extrapolation for efficient track-cluster matching

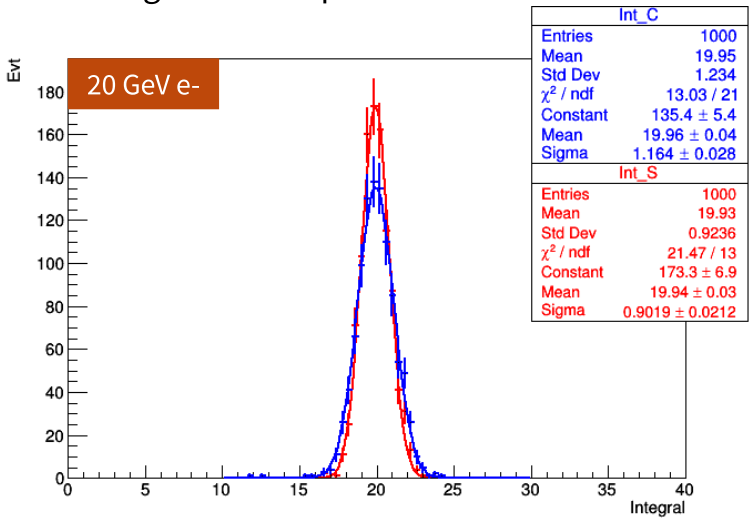
- Note: IDEA option 2 has a solenoid between the ECAL and HCAL
  (the issue is delegated from the track-cluster matching to the clustering)

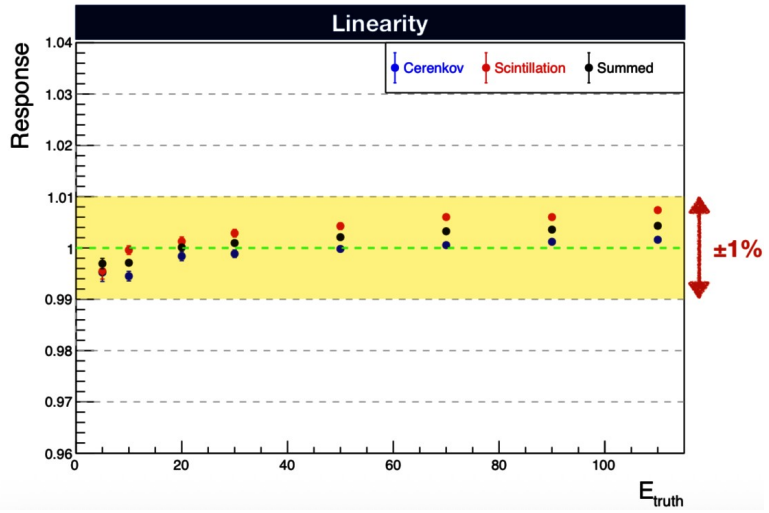


Calorimetry

Equalization of the Dual-readout calorimeter

- DRC has the same EM response for the scintillation and Cherenkov channels
- Integrated ADC of each channel is equalized to have the exact energy response for the 20 GeV e-
  - Simulated with the "no material" in front of the calorimeter & “ideal” condition (no noisy cell above the threshold)
  - No need to calibrate/rescale with different energies as the DRC has a decent EM energy response linearity (and hadron calibration comes for free for the option 1)
- Assuming that the equalization constants are the same as a function of pseudorapidity





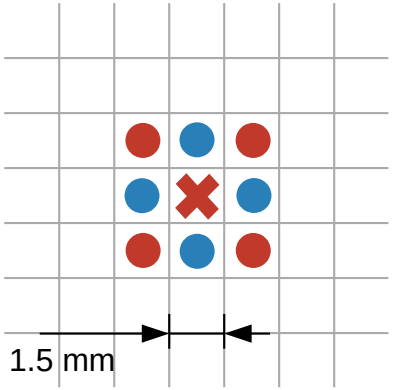
Clustering

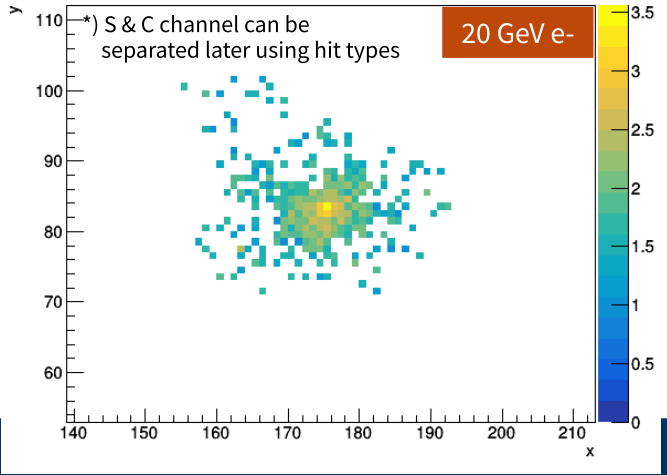
Topological clustering

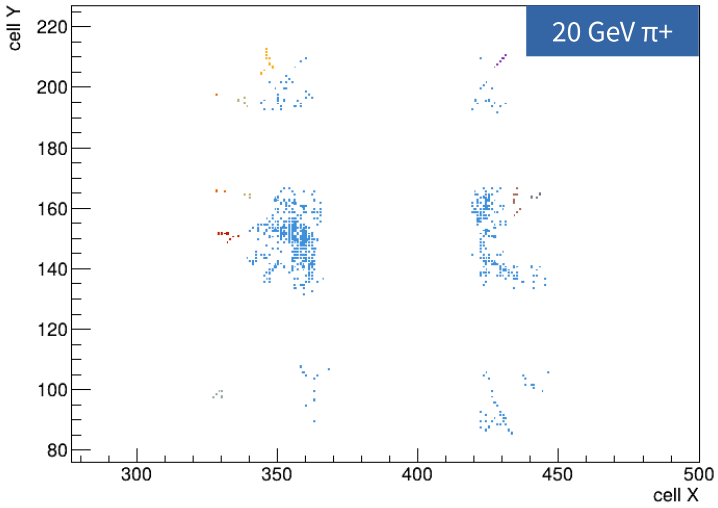
- Calo hits are clustered according to their topological neighborhood
  - Take calo hits more than 4σ of the noise level as the seeds
  - Grow clusters with the cells greater than the 2σ of the noise level
  - Incorporate any neighboring calo hit (0σ) in the last iteration
- To maintain uniform sampling fraction across the calorimeter, short fibers are inserted in the wing region of the trapezoid tower

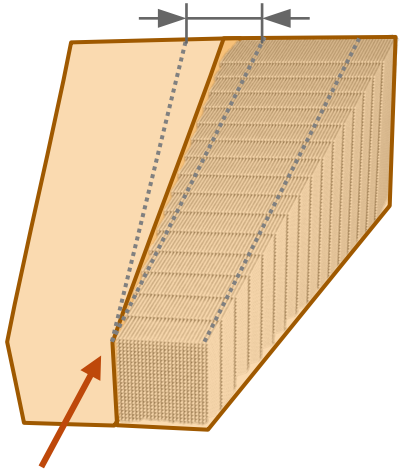
  → the neighborhood is defined based on the front-end of the calorimeter

  (calo hits in the wing region are always considered to be the neighbor)









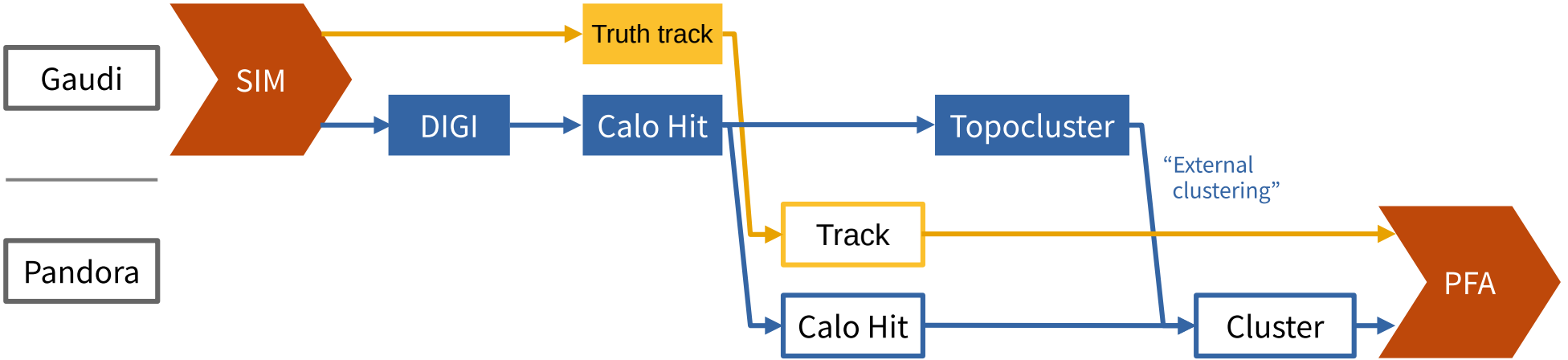
Reconstruction

Pre-PFA workflow

- Topological clustering heavily relies on the cell geometry – (basic) clustering is handled outside Pandora

- `pandora::Cluster` objects are created by the "External clustering" Pandora algorithm [git] based on the topoclusters
- Link `pandora::CaloHit` to the cluster according to the link from `edm4hep::Cluster` to `edm4hep::CalorimeterHit`

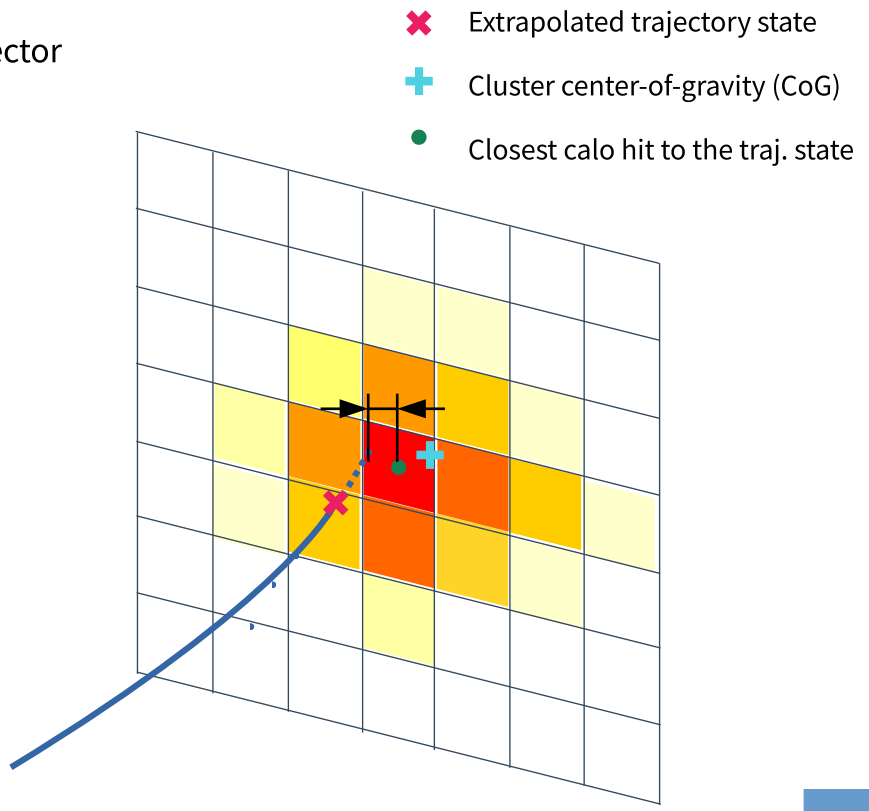
- Created `pandora::Track` and `pandora::Cluster` (and `pandora::CaloHit` depending on the possibility of reclustering) become base ingredient for the subsequent particle flow algorithms (PFAs)



Track-cluster matching

“LCContent” track-cluster association

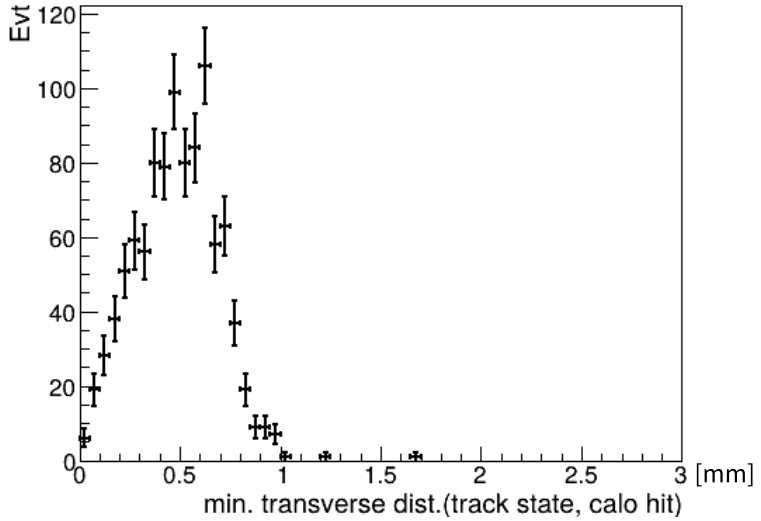
- The full PFA chain of LCContent is not applicable for the IDEA detector
  - Significantly different detector concept
  - Cherry-pick or build new PFAs based on the needs
- Track-cluster matching is one of the most ubiquitous algorithm commonly used for various detector concepts
  - very few assumptions on how the detector works
  - purely relies on the geometry
- Track-cluster association in LCContent is also applicable to IDEA
  - simple cut on the transverse distance from the extrapolated trajectory state to the closest calo hit
  - parallel distance cut also exists but less important for the calorimeter without logitudinal segmentation
- cut on 3mm (twice of the cell-to-cell distance)

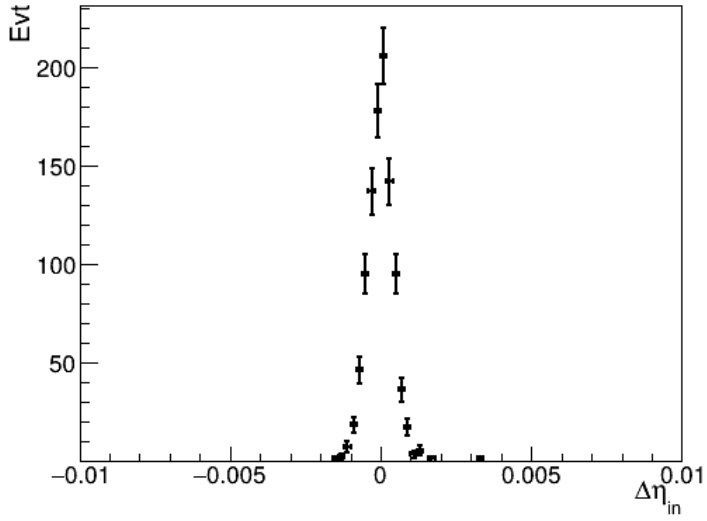


Track-cluster matching

Figures of merit

- Portrayed several figures of merit to understand the goodness of track-cluster matching
  - 10 GeV e- gun, uniform distribution within -0.88 < |η| < 0.88 (barrel)
  - Overall track-cluster matching efficiency is about 99% (subject of further studies for lower p_T particles)
- The transverse distance from the trajectory state to the closest calo hit is mostly within a unit cell size (1.5 mm)
- The Δη between the trajectory state and the cluster CoG ($\Delta\eta_{in}$) shows a sharp Gaussian with μ=0

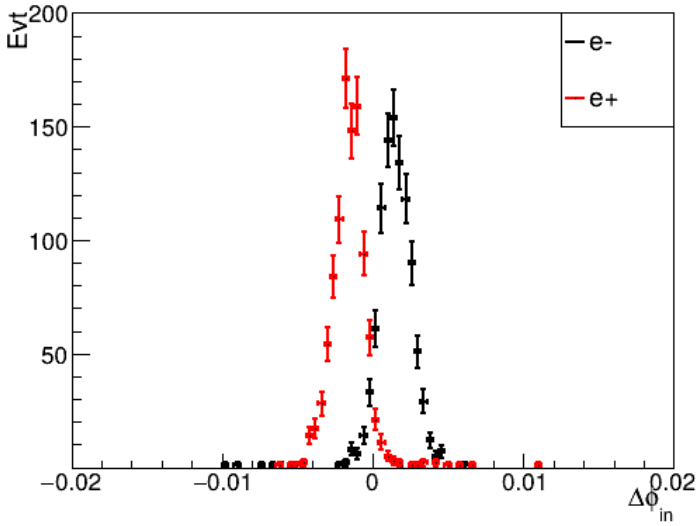


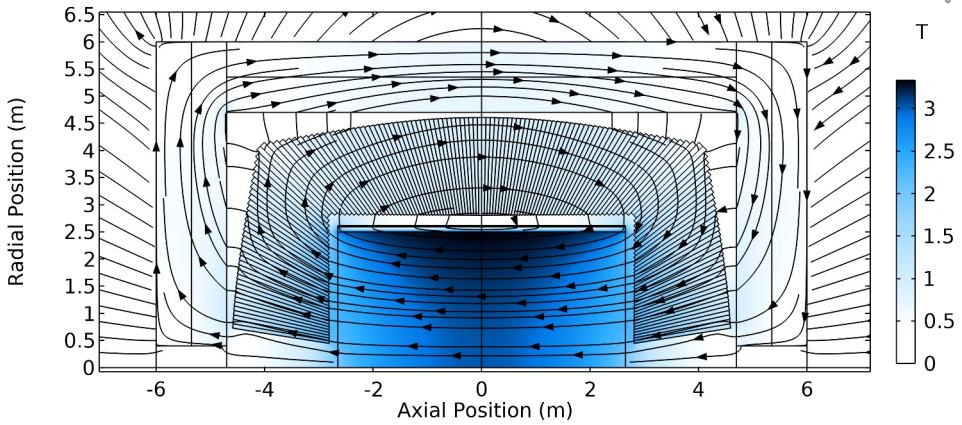


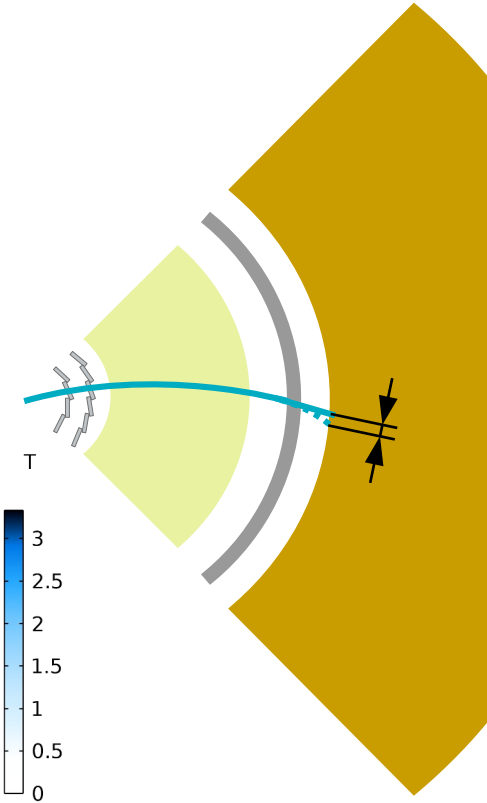
Track-cluster matching

Figures of merit

- However, the Δϕ between the trajectory state and the cluster CoG ($\Delta\phi_{in}$) is shifted depending on the charge due to the oversimplified trajectory extrapolation
  → needs more detailed extrapolation based on the local magnetic field
  (field between the solenoid and the calorimeter is likely close to 0)



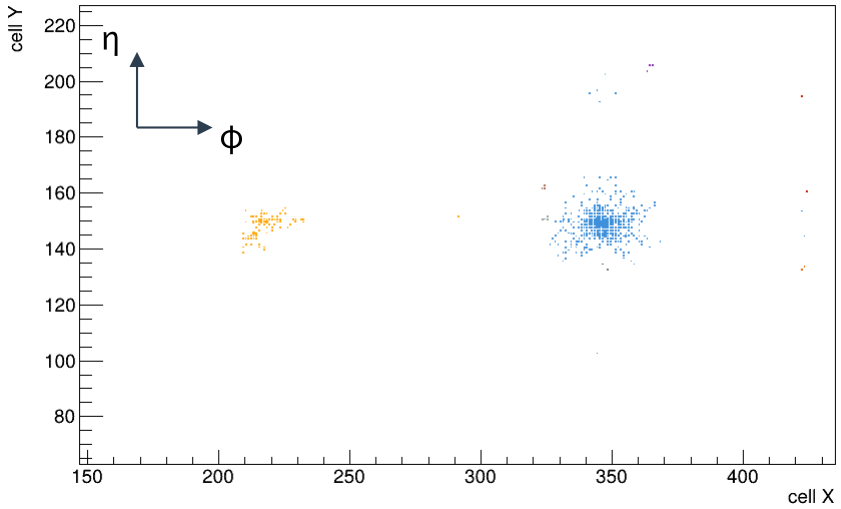


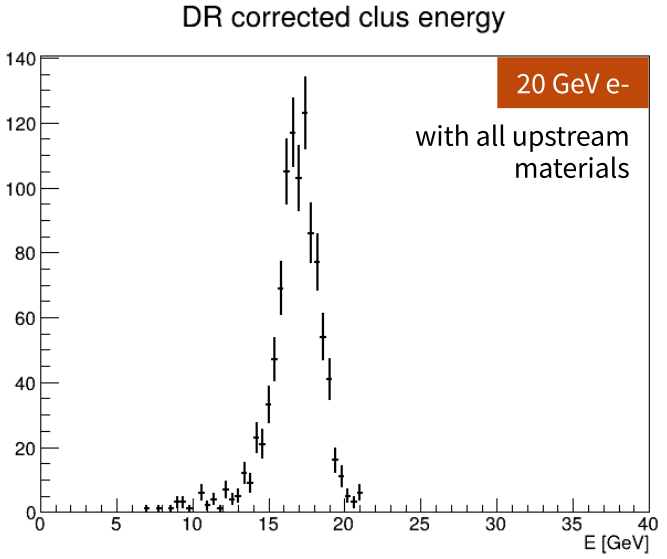


Future plans

Brem & radiation loss recovery

- Despite ultra-thin upstream subdetectors (and the solenoid for option 1), the electron radiation energy loss is not negligible
  - A single cluster energy shows a radiative tail on the low-energy side
  - Most of such events have an additional cluster along the phi direction
- Expect better electron energy description via reclustering/superclustering brem photons (and the energy loss correction)
- Tangents from the electron track would be beneficial to locate brem photons





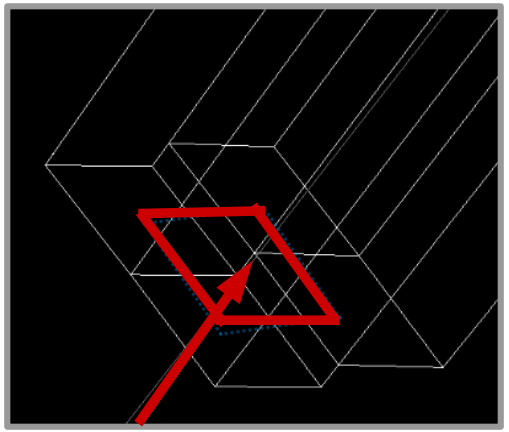
Future plans

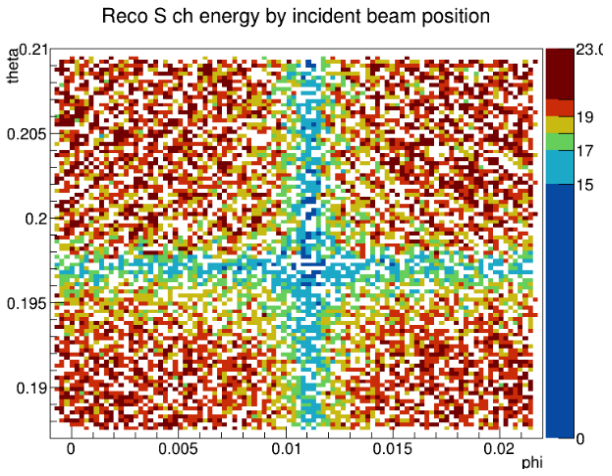
Energy regression

- Despite efforts to maintain a uniform sampling fraction in the wing region of the (projective) tower, the energy response as a function of the particle's incident position is not perfectly identical

- There exists a gap (or dead material) between the modules, lowering the sampling fraction at the edge of the tower
  - Shot a 20 GeV e- gun of a uniform distribution in the rectangular region at the middle of the 4 towers
  - lower reconstructed energy in the edge region compared to the center of the tower

- An energy regression algorithm that takes the shower shape, cluster position w.r.t. the module can improve the energy resolution



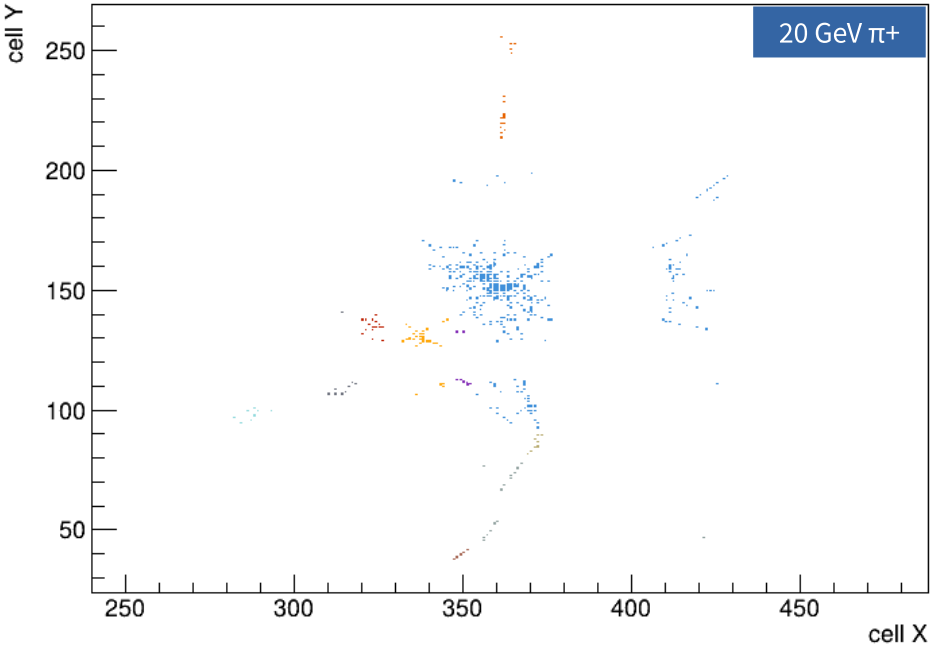


Future plans

Reclustering satellite clusters

- Extreme granularity of the calorimeter gives an excellent opportunity to resolve incoming particles
- But also adds a challenge for clustering secondary particles generated by the hadronic shower
  - Energy deposits of the secondary (hadronic) particle can be topologically detached from the core of the shower

    (in contrast to the EM shower where the energy deposits are always continuous)
- A reclustering algorithm would be needed to incorporate all energy deposits of a hadronic shower
- Despite satellite clusters, it's better to have initial clusters as granular as possible
  - Better π^0 identification efficiency at higher energy
  - Easier to merge clusters than splitting



Future plans

C/S and the cluster-level PID

- The dual-readout calorimeter has a unique feature to distinguish EM and hadronic shower

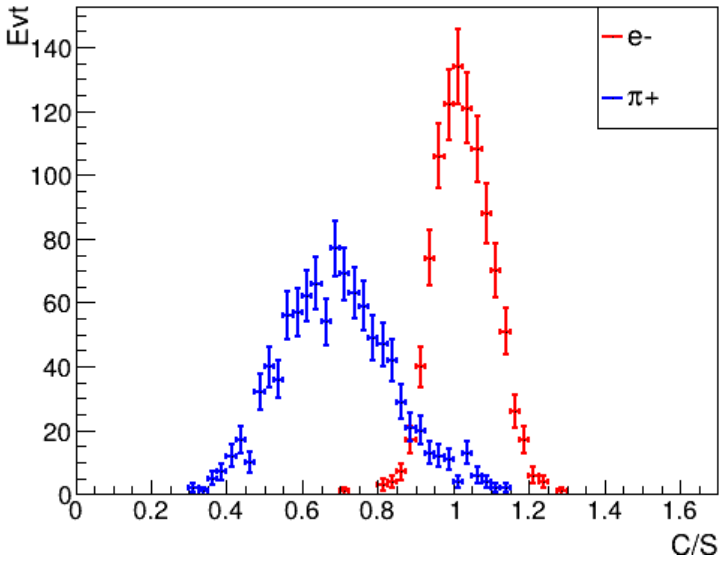
- A hadronic shower is likely to have less response in the Cherenkov channel compared to the scintillation channel

  (mainly sensitive to the light and relativistic charged particles, i.e. $e^{\pm}$)

  - The ratio between the scintillation and Cherenkov channels gives excellent separation between the EM & hadronic shower
  - Compensates for the lack of longitudinal segmentation

- can be helpful to reduce the confusion between particle types and to trigger the right downstream PFA

  (e.g. reclustering, dual-readout correction, etc)



Pattern recognition via timing

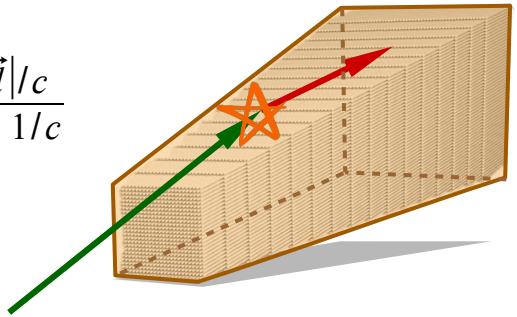
Waveform analysis

- Unsegmented calorimeter fully depends on the timing to reconstruct longitudinal shower shape
- A SiPM yields exponentially decaying waveform to 1 photon
- A Fourier analysis on the waveform can be used to mitigate exponential tail while preserving time translation & amplitude information
- With 10 GHz sampling rate, a unit time bin (100 ps) corresponds to about 4 cm in the longitudinal direction
- More details from the talk at [CALOR2022]

Deposit position $(\vec{x})$ Photon propagation $(\vec{k})$

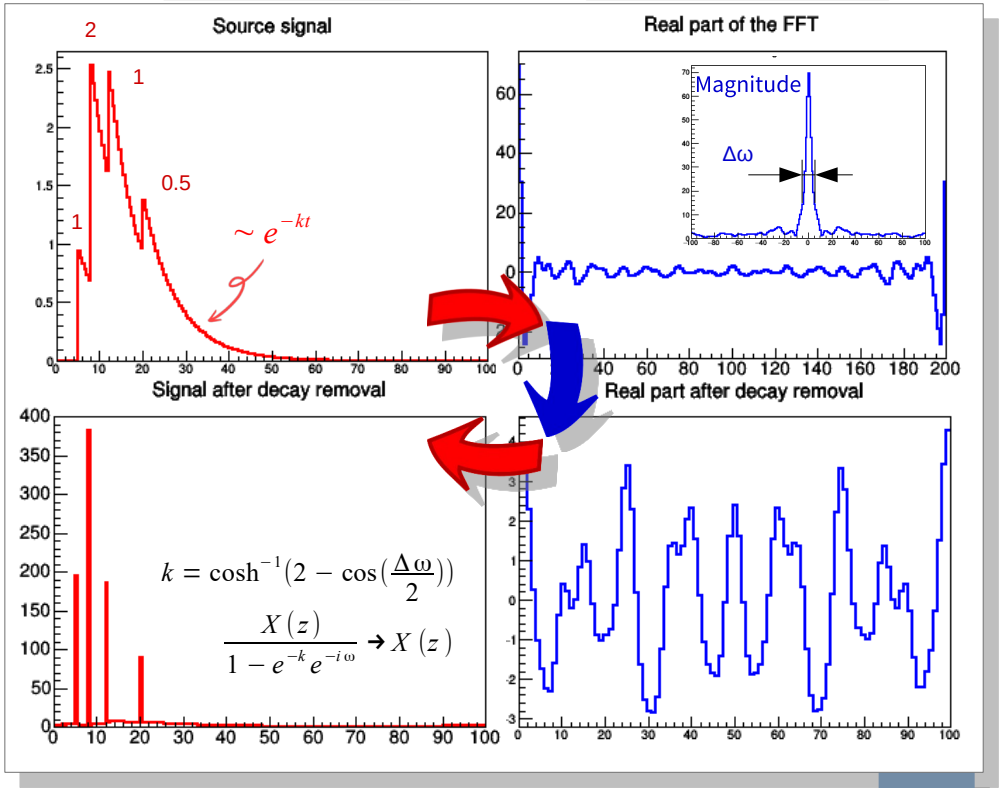
$$t = \frac{|\vec{x}|}{c} + \frac{|\vec{k}|}{v} \qquad |\vec{k}| \simeq \frac{t - |\vec{l}|/c}{1/v - 1/c}$$

$$\vec{x} \simeq \vec{l} - \frac{t - |\vec{l}|/c}{1/v - 1/c}\hat{k}$$

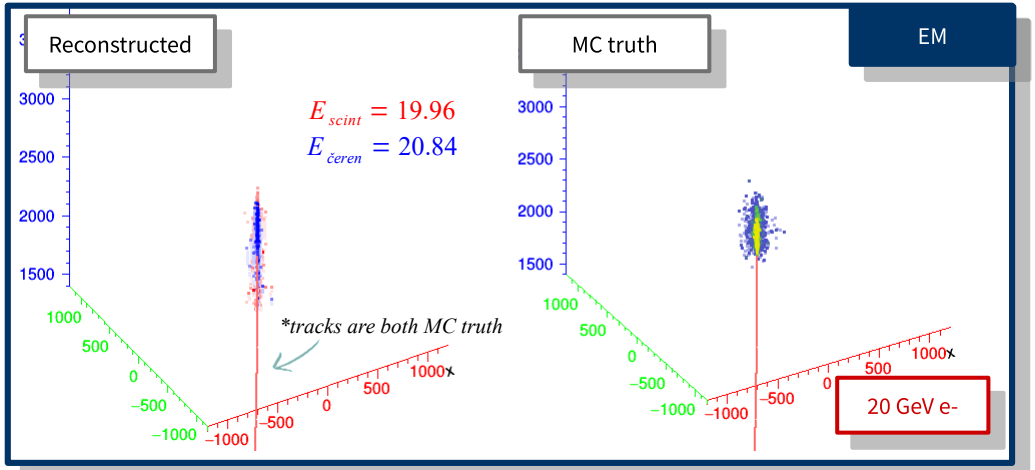


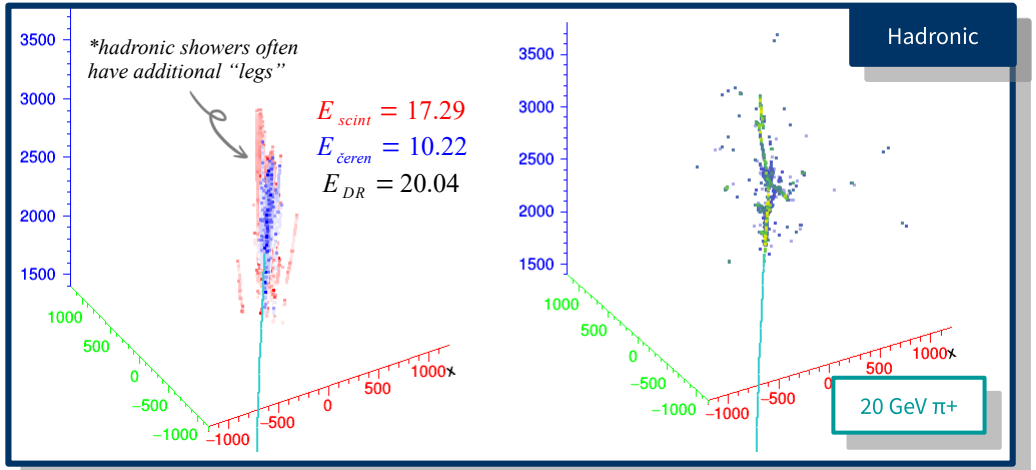
Time domain

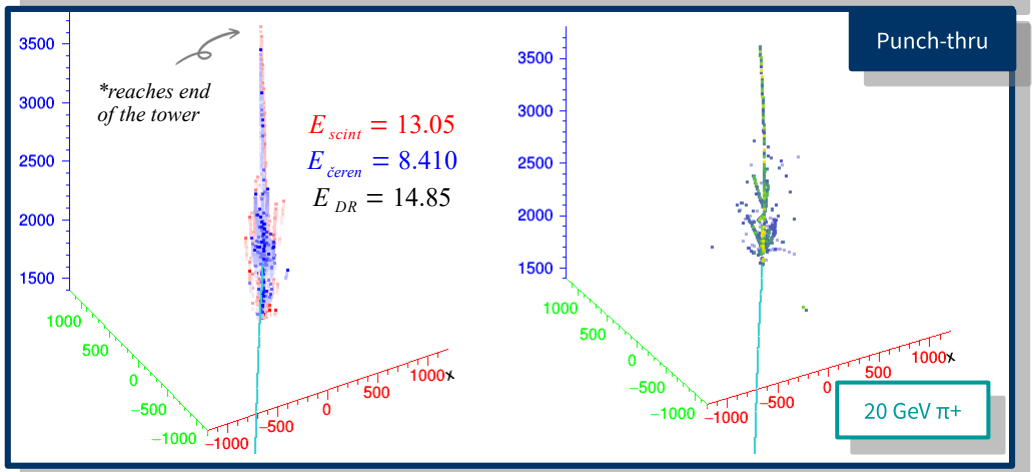
Frequency domain

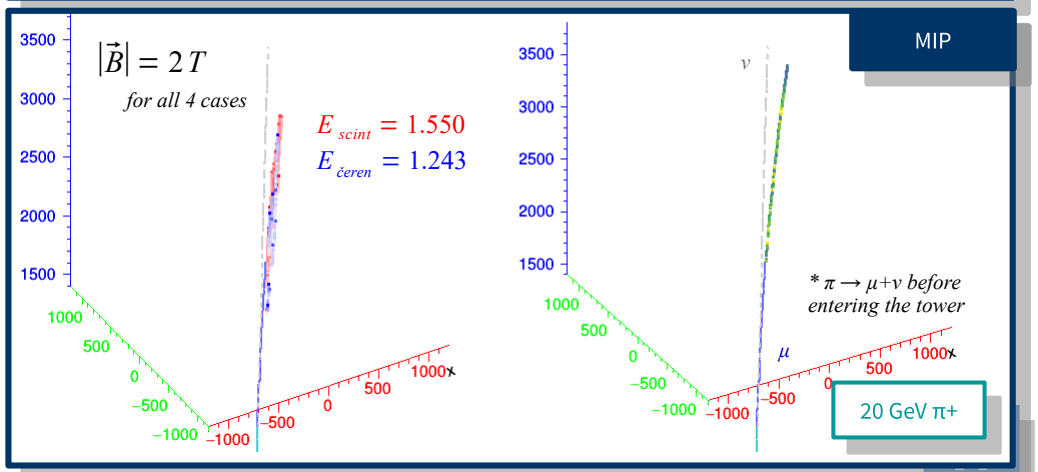


Pattern recognition via timing









Summary

Pandora SDK and the IDEA detector

- Utilization of the Pandora SDK for the IDEA detector has just started

- Many things need to be done
  - Realistic tracking & dN/dx PID feature to exploit the potential of the drift chamber
  - Needs input to optimize the crystal dual-readout calorimeter

- Clusters are built outside using the topological clustering algorithm
  (and copied to Pandora by the external clustering algorithm)

- Preliminary track-cluster association has been done using the existing algorithm in LCContent
- Many of the other PFA blocks are to be explored (brem recovery, reclustering, energy regression, etc)
  - Extend to more complicated cases (e, μ, γ, $\pi^{\pm}$, K^{0}_{L}, and jets)
  - … but keep it as simple as possible to allow flexibility for multiple detector options

- Full waveform analysis is an interesting possibility to improve pattern recognition with limited longitudinal granularity
  (caveat: the proof of concept needs to be demonstrated with data)



Backups

Calo hit creation

```
for (const auto& hit : inputCaloHits) {
  // create Pandora calo hit
  PandoraApi::CaloHit::Parameters caloHitParameters;
  // see PandoraSDK/include/Pandora/ObjectCreation.h for the full list
  auto pos = hit.getPosition();
  caloHitParameters.m_positionVector = pandora::InputCartesianVector(pandora::CartesianVector(pos.x, pos.y, pos.z));
  caloHitParameters.m_cellGeometry = pandora::InputCellGeometry(pandora::CellGeometry::POINTING);
  caloHitParameters.m_cellSize0 = 1.5; // mm (edm4hep unit) FIXME make configurable
  caloHitParameters.m_cellSize1 = 1.5; // mm (edm4hep unit) FIXME make configurable
  caloHitParameters.m_time = pandora::InputFloat(hit.getTime());
  caloHitParameters.m_inputEnergy = pandora::InputFloat(hit.getEnergy());
  caloHitParameters.m_hitType = pandora::HitType::ECAL; // TODO push scint & Cheren calo hit types to PandoraSDK
  caloHitParameters.m_hitRegion = pandora::HitRegion::SINGLE_REGION; // TODO identify barrel or endcap

  // dummy parameters
  caloHitParameters.m_isDigital = pandora::InputBool(false);
  caloHitParameters.m_layer = 0; // always zero for DRC
  caloHitParameters.m_expectedDirection = caloHitParameters.m_positionVector.Get().GetUnitVector();
  caloHitParameters.m_cellNormalVector = caloHitParameters.m_positionVector.Get().GetUnitVector();
  caloHitParameters.m_cellThickness = 2000.; // mm (edm4hep unit) FIXME make configurable
  caloHitParameters.m_nCellRadiationLengths = 100.; // FIXME make configurable
  caloHitParameters.m_nCellInteractionLengths = 8; // FIXME make configurable
  caloHitParameters.m_mipEquivalentEnergy = 0.1; // GeV (edm4hep unit) FIXME make configurable
  caloHitParameters.m_electromagneticEnergy = caloHitParameters.m_inputEnergy; // TODO this parameter doesn't make sense at all for DRC
  caloHitParameters.m_hadronicEnergy = 0.; // TODO this parameter doesn't make sense at all for DRC (for now assume pure EM)
  caloHitParameters.m_isInOuterSamplingLayer = false;

  // address to edm4hep calo hit
  caloHitParameters.m_pParentAddress = pandora::InputAddress(&hit);

  PANDORA_THROW_RESULT_IF(pandora::STATUS_CODE_SUCCESS, !=,
                          PandoraApi::CaloHit::Create(m_pandora, caloHitParameters));
} // hits
```

Track-cluster association

```
<pandora>
    <!-- GLOBAL SETTINGS -->
    <IsMonitoringEnabled>true</IsMonitoringEnabled>
    <ShouldDisplayAlgorithmInfo>true</ShouldDisplayAlgorithmInfo>
    <ShouldCollapseMCParticlesToPfoTarget>false</ShouldCollapseMCParticlesToPfoTarget>

    <algorithm type = "ExternalClustering">
        <FlagClustersAsPhotons>false</FlagClustersAsPhotons>
    </algorithm>

    <algorithm type = "TrackClusterAssociation">
        <LowEnergyCut>0.5</LowEnergyCut>
        <MaxTrackClusterDistance>25.</MaxTrackClusterDistance>
        <MaxSearchLayer>0</MaxSearchLayer>
        <ParallelDistanceCut>500.</ParallelDistanceCut>
        <MinTrackClusterCosAngle>0.</MinTrackClusterCosAngle>
    </algorithm>

    <algorithm type = "CreatePfo">
        <OutputPfoListName>PfoIdea</OutputPfoListName>
    </algorithm>

</pandora>
```

Dual-readout calorimeter

Dual-readout calorimetry

- The major difficulty of measuring energy of hadronic showers comes fro the fluctuation of EM fraction of a shower, f_em
- f_em can be measured by implementing two different channels with different h/e response in a calorimeter

$$S = E\left[f_{em} + \left(\frac{h}{e}\right)_S (1 - f_{em})\right],$$

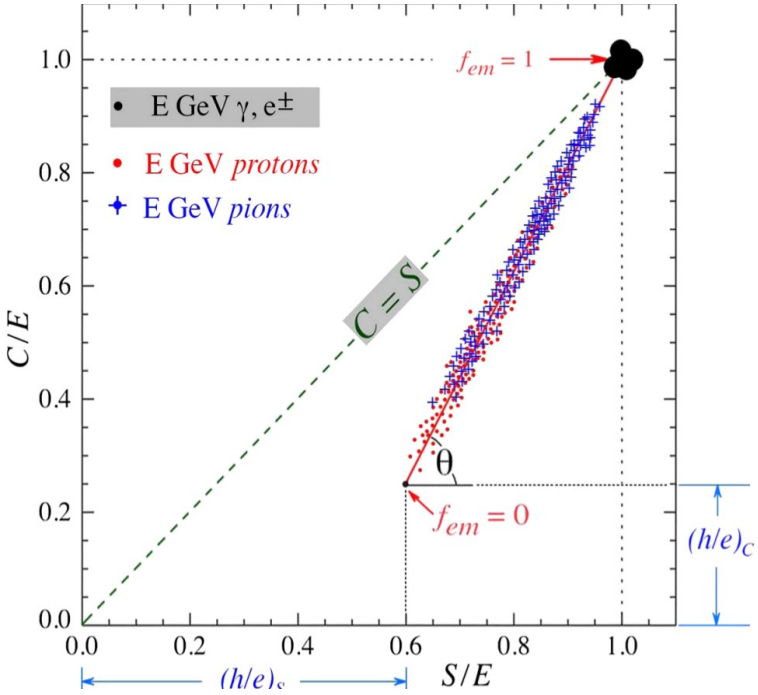
$$C = E\left[f_{em} + \left(\frac{h}{e}\right)_C (1 - f_{em})\right]$$

$$f_{em} = \frac{(h/e)_C - (C/S)(h/e)_S}{(C/S)[1-(h/e)_S] - [1-(h/e)_C]}$$

$$\cot\theta = \frac{1-(h/e)_S}{1-(h/e)_C} \equiv \chi,$$

$$E = \frac{S - \chi C}{1 - \chi}$$

- Excellent energy resolution for hadrons can be achieved by measuring f_em and correcting the measurement event-by-event
- Dual-readout fiber-sampling calorimeter is a key element of the IDEA detector concepts



Energy measured from scintillation channel vs Cerenkov channel for EM particle, π & p

Calibration

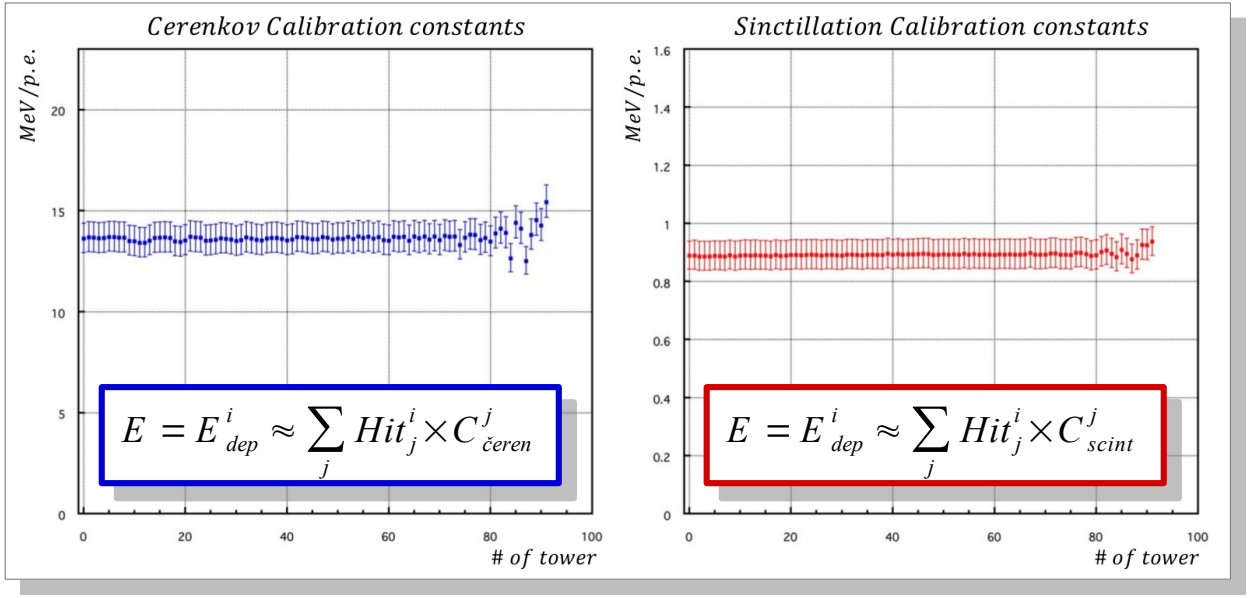
Calibration using 20 GeV e-

- Measure Energy deposit, scintillation p.e. & Čerenkov p.e. at i-th tower (0th - 91st)
- Energy can be expressed as a linear combination with simulations of 92 towers
  → Estimate calibration constants

- Uniform calibration constants as a function of the tower number

$$Energy = \sum_{i=0}^{92} Hit_{i^{th}\ tower} \times Calibration\ constant^{i^{th}\ tower}$$

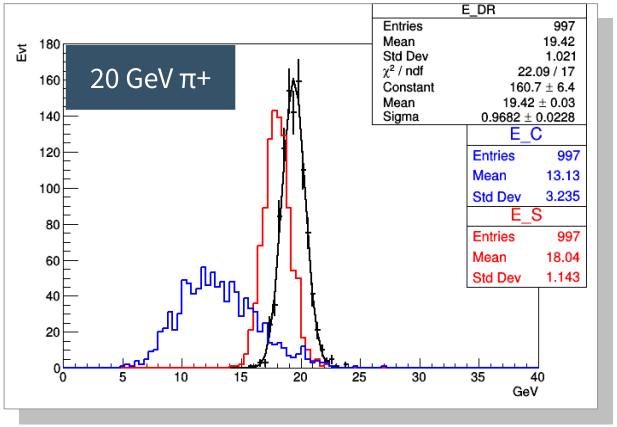
$$\Rightarrow \begin{bmatrix} E_{dep}^{0} \\ E_{dep}^{1} \\ \vdots \\ E_{dep}^{90} \\ E_{dep}^{91} \end{bmatrix} = \begin{bmatrix} Hit_0^0 & Hit_0^0 & \cdots & Hit_{90}^0 & Hit_{91}^0 \\ Hit_0^1 & Hit_1^1 & & Hit_{90}^1 & Hit_{91}^1 \\ \vdots & & \ddots & \vdots & \\ Hit_0^{90} & Hit_1^{90} & & Hit_{90}^{90} & Hit_{91}^{90} \\ Hit_0^{91} & Hit_1^{91} & \cdots & Hit_{90}^{91} & Hit_{91}^{91} \end{bmatrix} \begin{bmatrix} C^0 \\ C^1 \\ \vdots \\ C^{90} \\ C^{91} \end{bmatrix}$$

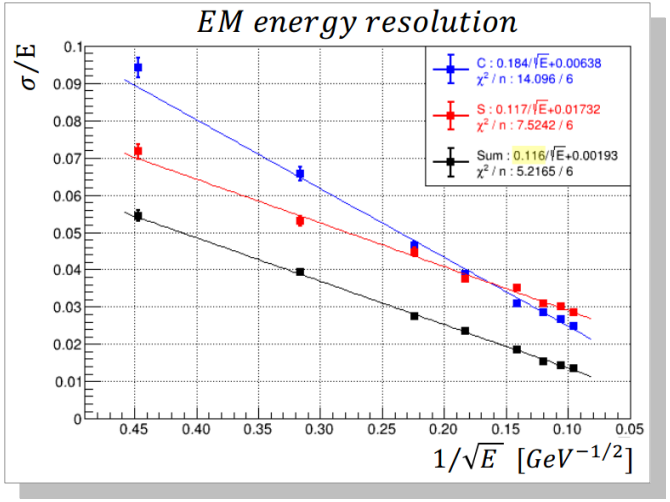


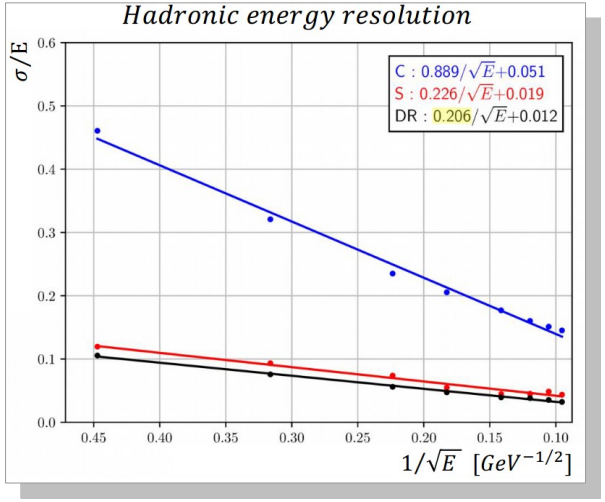
Energy resolution

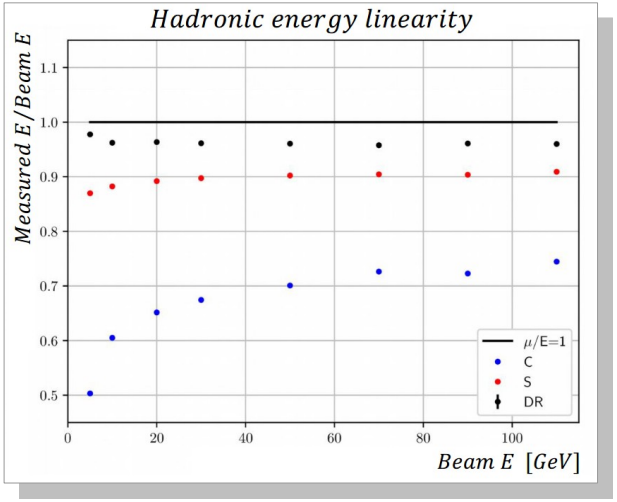
Estimation of energy resolution with GEANT4

- GEANT4 shows excellent energy resolution for both EM & hadronic showers
  → 11.6%/√E (EM) 20.6%/√E (Hadronic)
- Moving forward to demonstrate energy resolution with the beam test data
  → details presented by INFN colleagues [link]





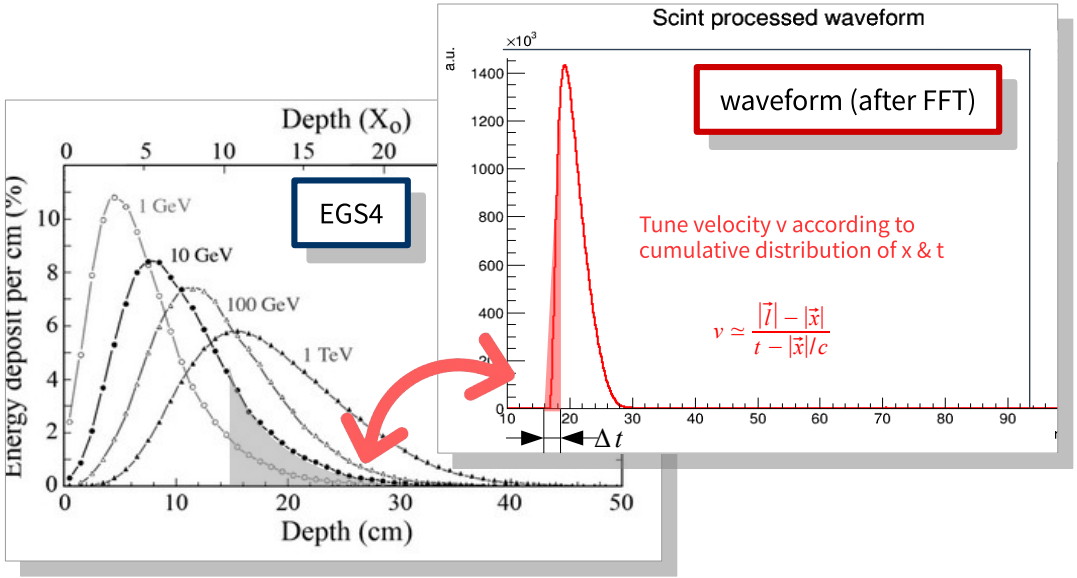


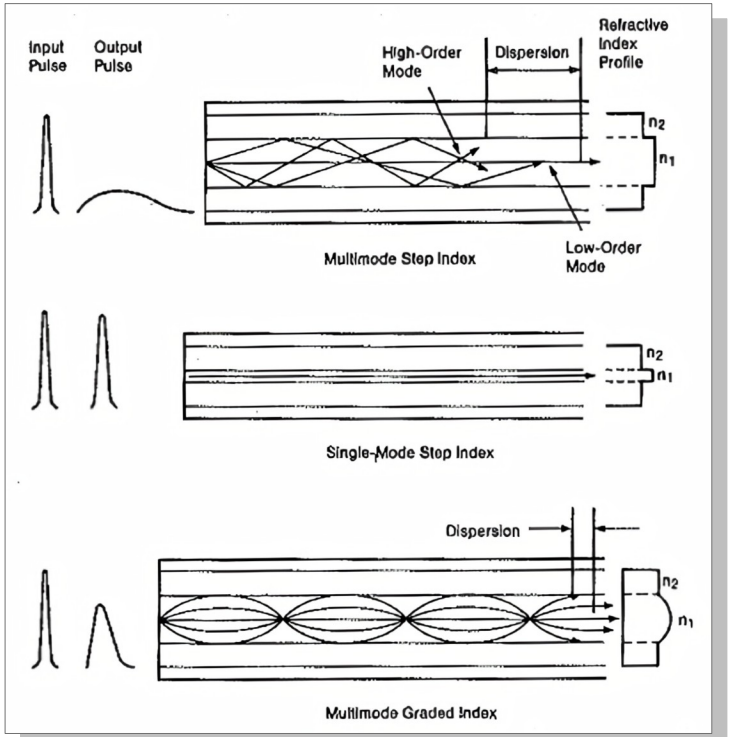


Longitudinal shower shape

Shower shape & timing – Dispersion

- Waveform is unlikely a shower shape even after FFT processing
- Late-component of the timing is dominated by the modal dispersion
- Mitigate dispersions by using slower phase velocity for late-components
  → Tune group velocity as a function of Δt using EM shower



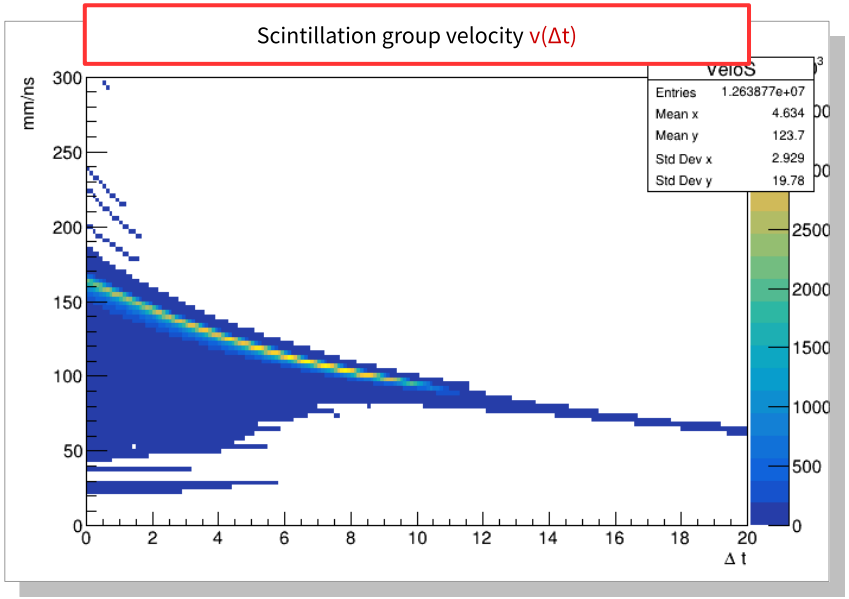


Modal dispersion

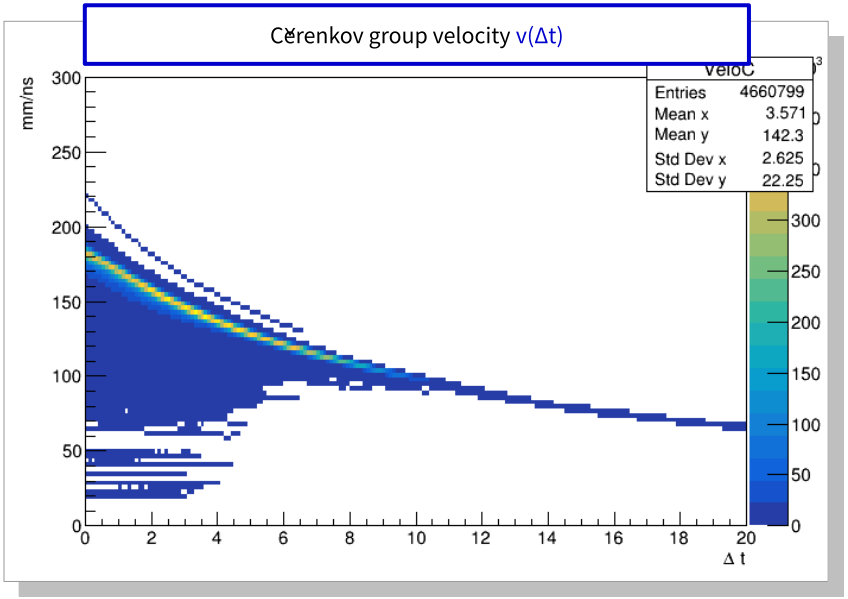
Group velocity modeling

- Assign slower group velocity for the late-components at $t = t_0 + \Delta t$ (ToA)
- Apply tuning according to cumulative distribution of dE/dx & dN/dt with 20 GeV e-
  → profile group velocity for every fiber by assuming the longitudinal shape (EM shower template)

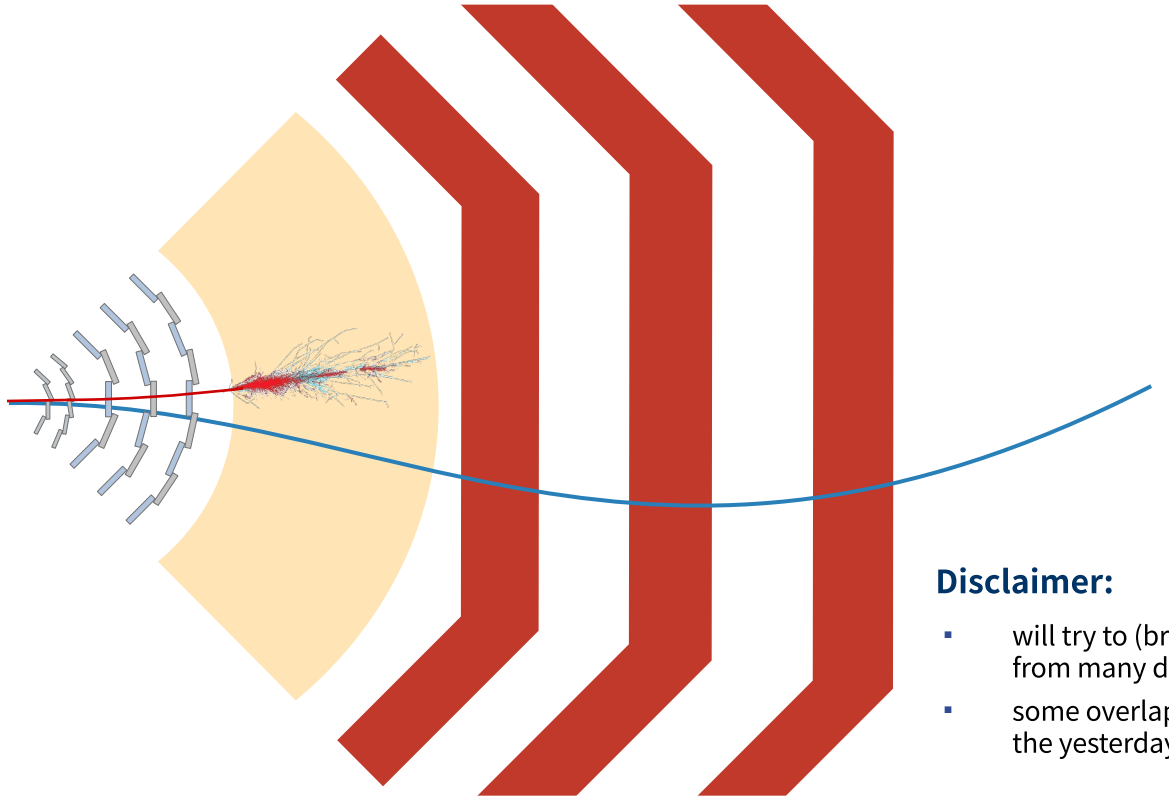
Scintillation group velocity v(Δt)



Čerenkov group velocity v(Δt)



Outline



(We will follow this order, but not strictly)

Disclaimer:

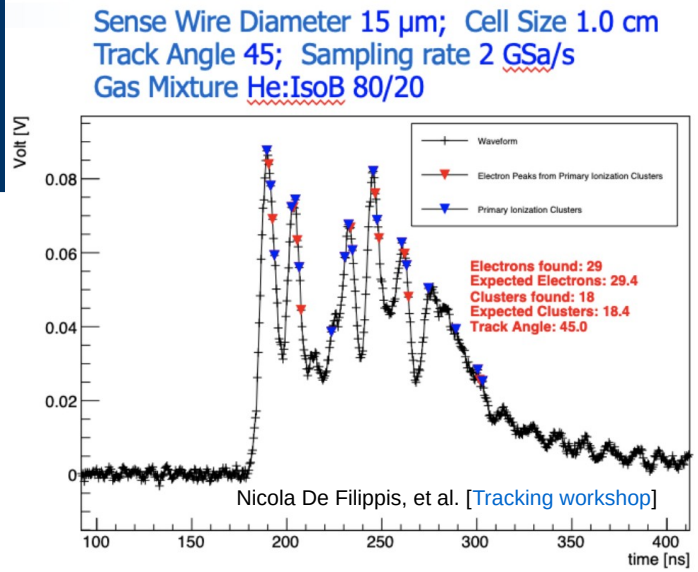
- will try to (briefly) cover various efforts from many different groups & people
- some overlaps with (and more details in) the yesterday's & today's PED sessions

Drift chamber & Vtx digitizers

Drift chamber digitizer

- Works ongoing to implement the IDEA drift chamber digitization [Fabrizio's talk]
  - A (unit) pulse is constructed using templates derived from the experiments
  - Summing up pulses generated w.r.t. sim hits would yield an analog waveform

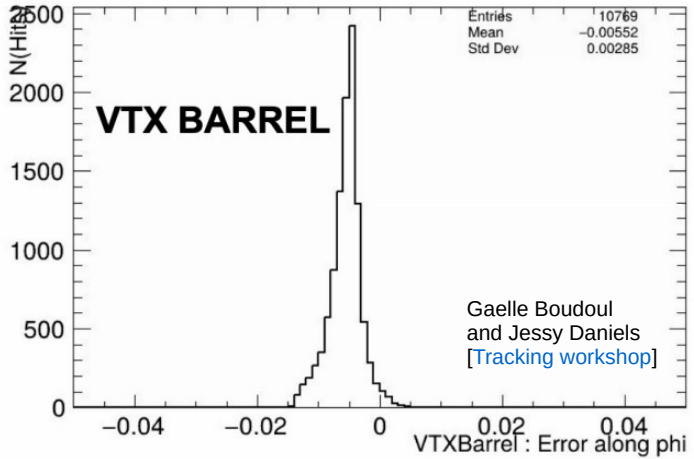
  → apply cluster counting algo and find the # of ionization clusters (and dN/dx)



Nicola De Filippis, et al. [Tracking workshop]

Vertex detector digitizer

- A detailed pixel detector digi is being developed (`VTXDigitizerDetailed`)
  (more than a simple Gaussian smearing)
  - Instrumental effects (noise, thresholds, etc)
  - Also consider charge propagation and diffusion
- Plan to implement time resolution effect as well



Gaelle Boudoul and Jessy Daniels [Tracking workshop]

Distance btn sim hits and digitized hits

Track reconstruction

GenFit2

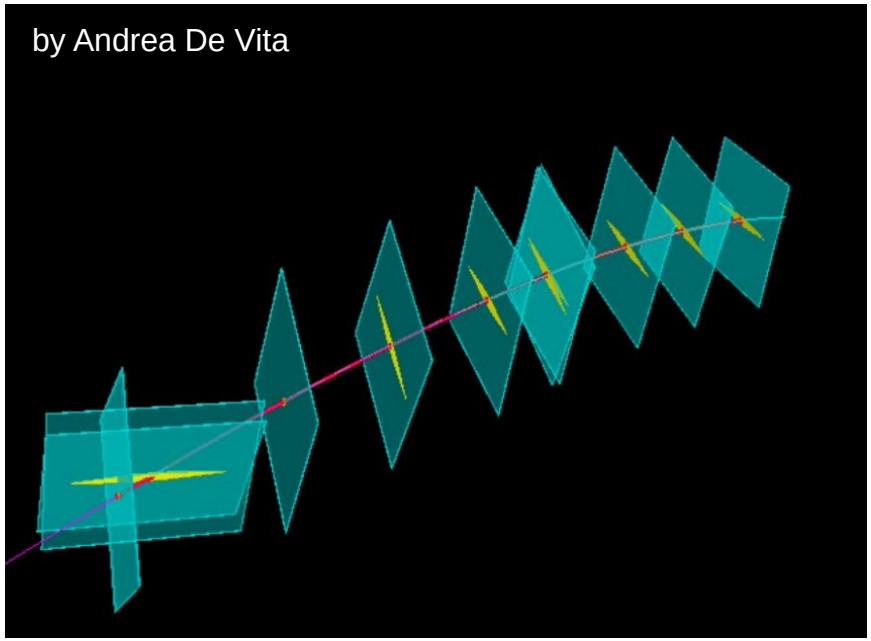
- A generic track-fitting toolkit based on the Kalman fitter(s)
- Developed in the context of Belle II [J. Phys.: Conf. Ser. 608 012042]
  - Provides experiment-independent track-fitting
  - and also graphic visualization of tracks & detectors

- Validated using the CLD detector
  (where we already have the conformal tracking)
- Implementation to the IDEA detector is ongoing
- To be implemented to k4RecTracker

- For preliminary studies, an even simpler workaround is available that creates tracks based on the MC-truth particle
  `(TracksFromGenParticles)`

2 GeV π+ event

(from left to right)
vtx endcap + inner & outer endcap

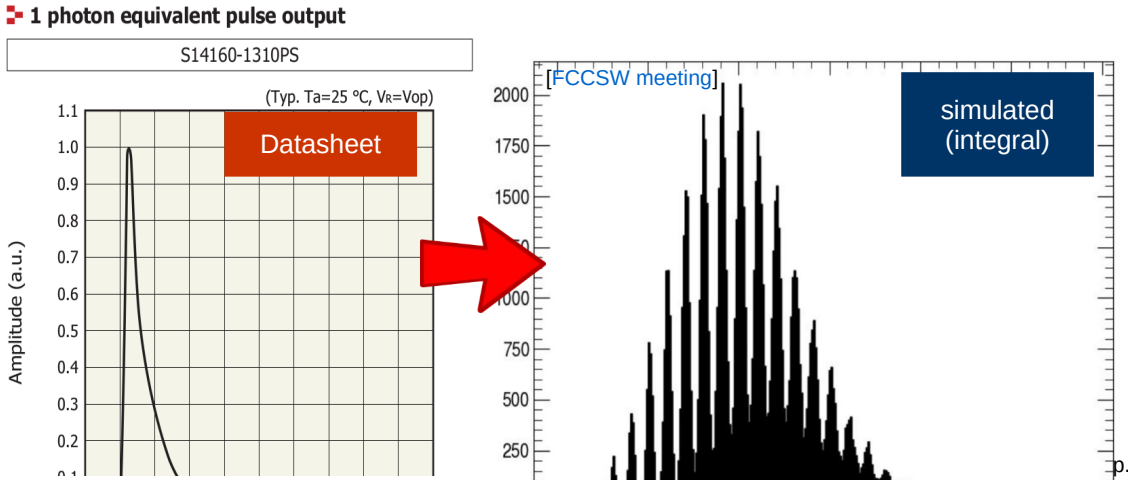


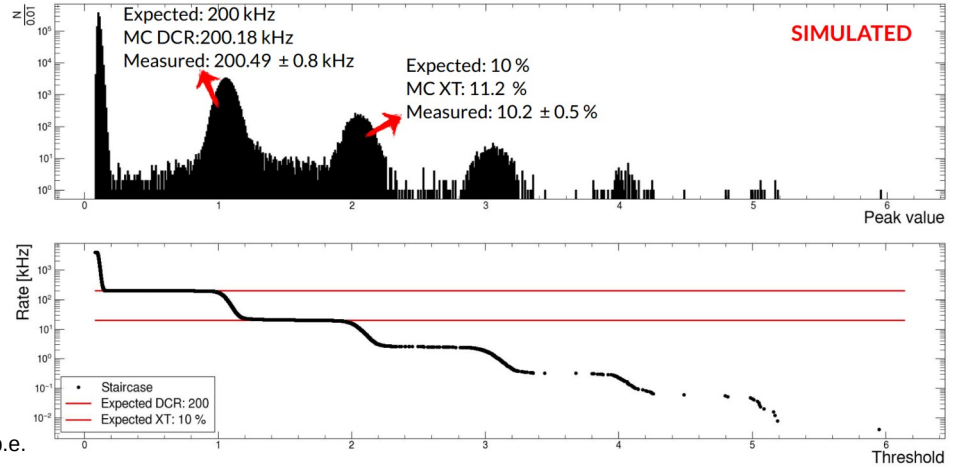
Calorimeter digitizers

SiPM emulation

- For detectors using SiPM, a dedicated emulation library "SimSiPM" has been developed
  → able to simulate the output waveform of SiPM based on parameterized inputs from the datasheet
  (dark counts, crosstalk, afterpulse, saturation, noise, …)
- Interfaced to k4RecCalorimeter recently [k4RecCalorimeter#145]

Example distributions for MIP signals
by Edoardo Proserpio and Romualdo Santoro





Clustering algorithms

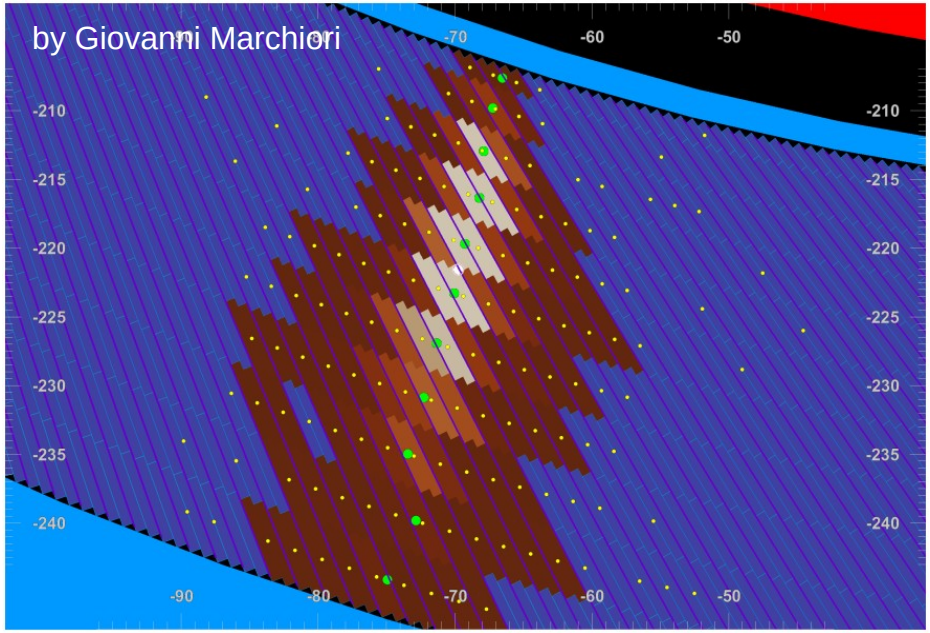
Sliding window clustering

- Finds a local maxima based on the sliding window of fixed size in θ×φ
  (caveat: assumes a θ×φ based detector segmentation)
- Implemented in the k4RecCalorimeter a while ago

Topological clustering

- Builds a cluster by grouping the topologically neighboring cells above a certain energy-to-noise ratio threshold
- The topological neighborhood can be defined from the user's side
  → easily applicable for various detector concepts/options

A topocluster for a 20 GeV e- w/ ALLEGRO barrel ECAL

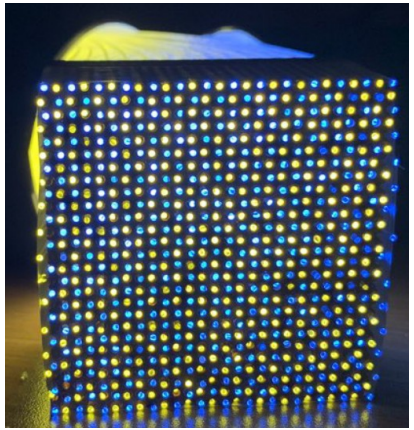


Clustering algorithms

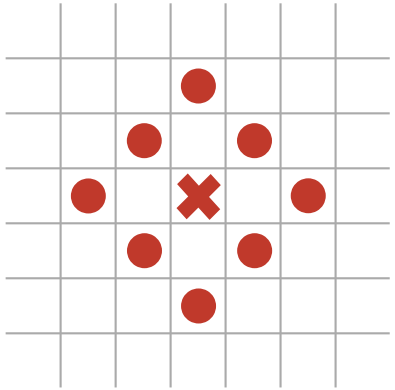
Topological clustering (cont'd)

- IDEA dual-readout calorimeter has two channels that can to be clustered independently
  (the topological neighborhood is different from the geometrical neighborhood)
  → a good example that the topological clustering can be applied regardless of the geometrical detail

Dual-readout calorimeter with scintillation and Cherenkov fibers



Definition of the topological neighborhood within the same channel

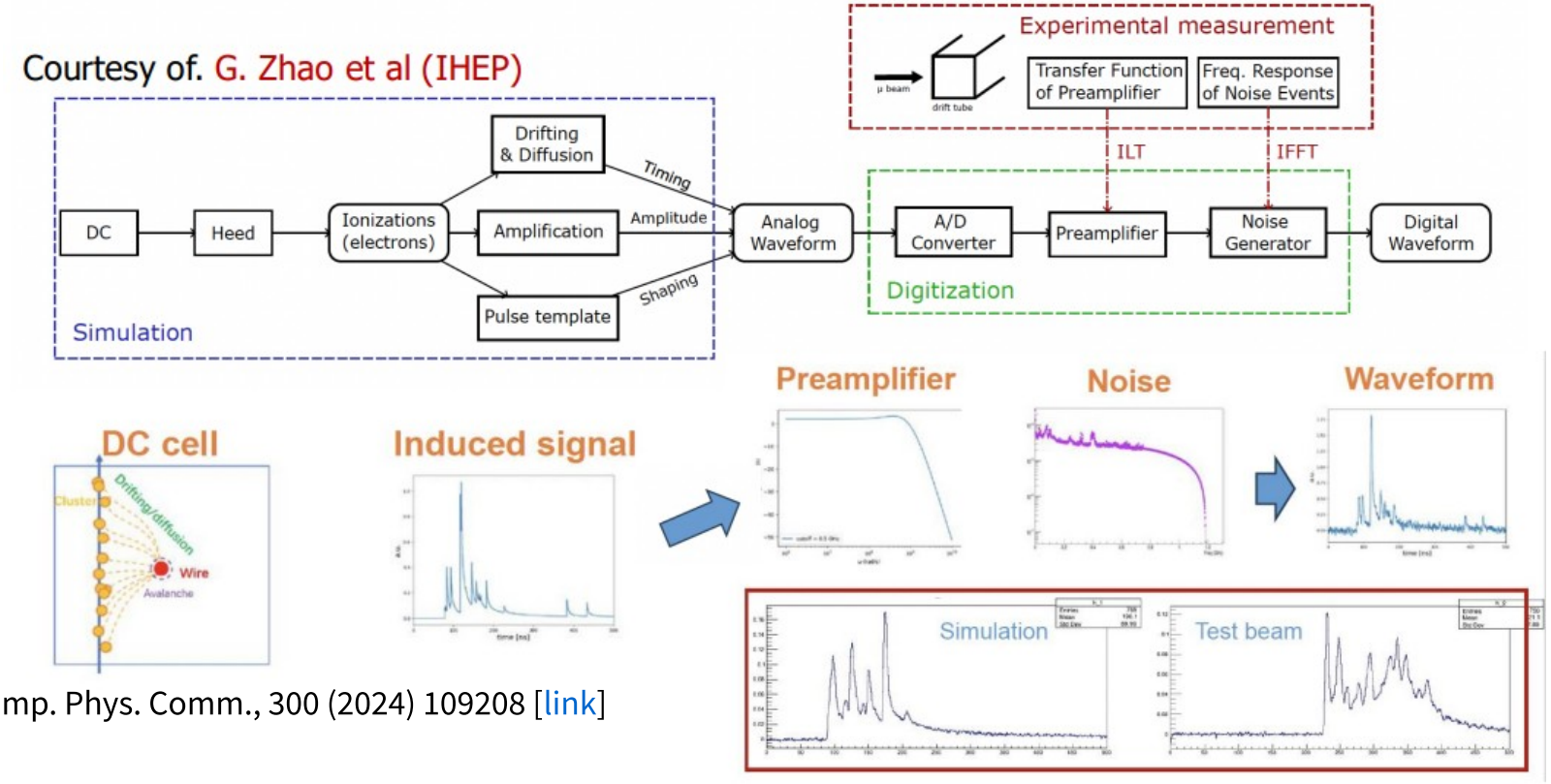


Two most energetic topoclusters for a 20 GeV π^0
(scintillation channel only)



z-axis scale = $\log_{10}$(E/MeV)

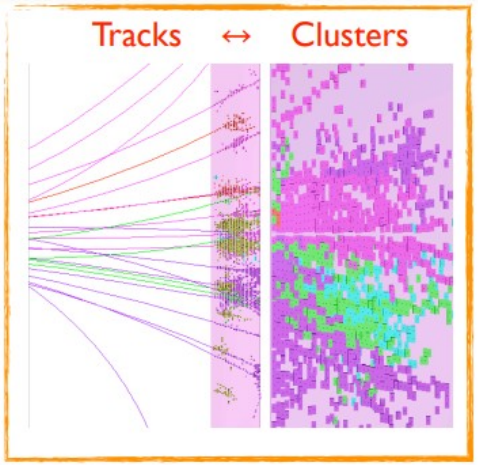
Drift chamber fullsim chain

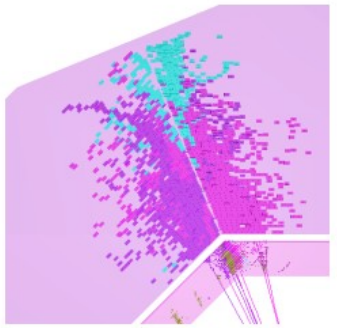


- Comp. Phys. Comm., 300 (2024) 109208 [link]

Pandora PFA

- Track-cluster association algs match cluster positions and directions with helix-projected track states at calorimeter.
- In very high-density jets, reach limit of "pure" particle flow: can't cleanly resolve neutral hadrons in hadronic showers.
- Identify pattern-recognition problems by looking for significant discrepancies between cluster E and track p.
- Choose to recluster: alter clustering parameters or change alg entirely until cluster splits and consistent E/p achieved.





After topological association

Compare E/p values to find problems



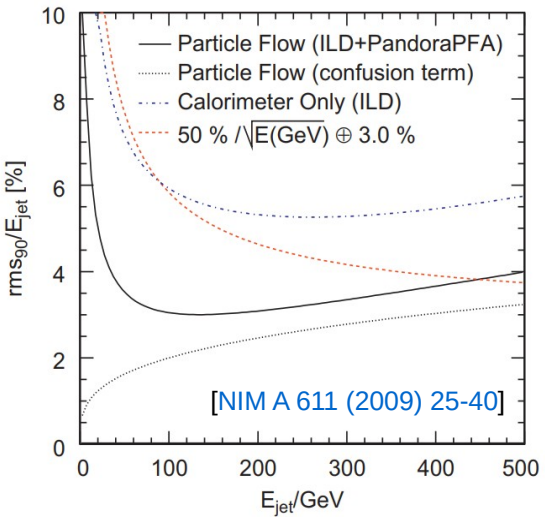
Find n absorbed into π- cluster

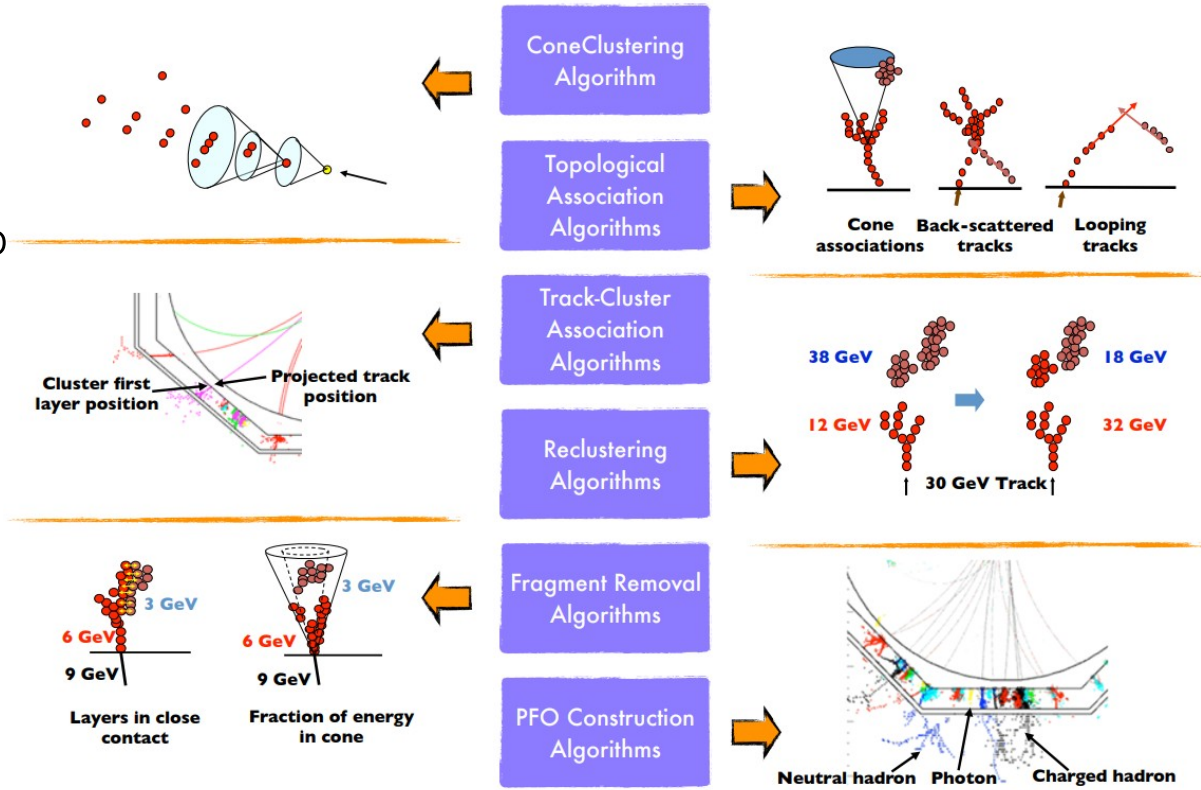
e.g. 45GeV track associated to 95GeV cluster:

identify and address clustering problem

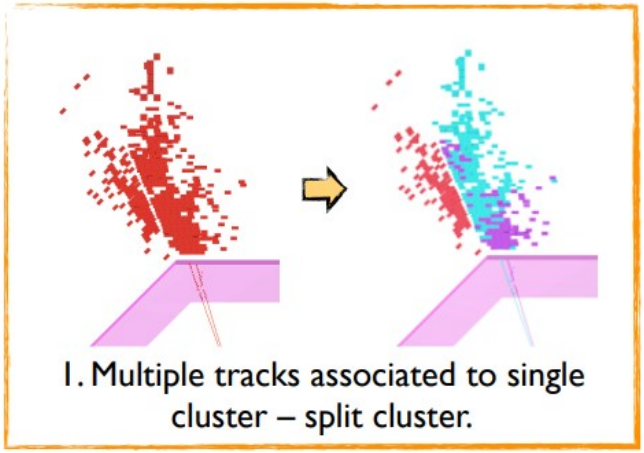
Pandora PFA

- Performs clustering, track-cluster association, reclustering, and PF object reconstruction for (but not limited to) PF calorimeters
- originally implemented with Marlin (iLCSoft) (currently works with k4MarlinWrapper)
- Available w/ CLD and being tested w/ ALLEGRO
- Work in progress to migrate to native Gaudi

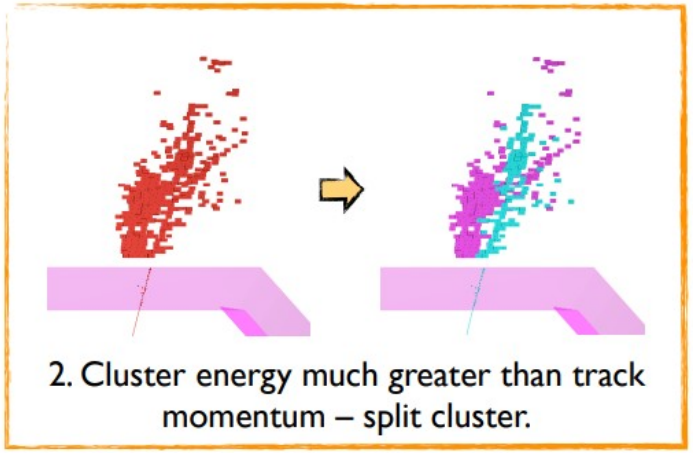




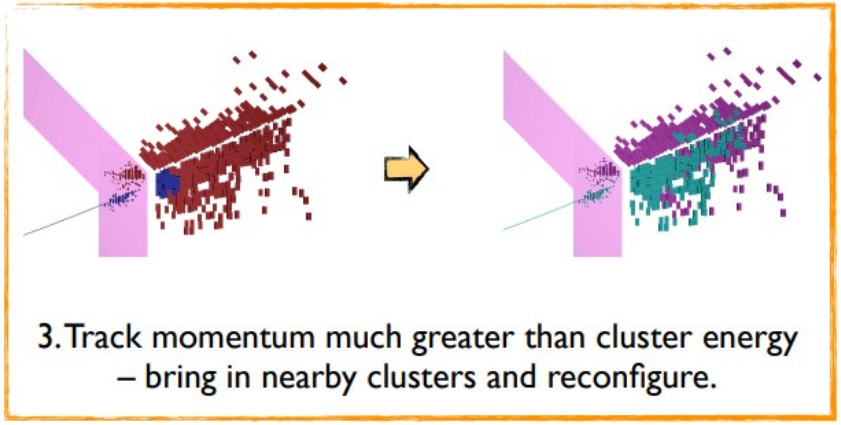
Pandora PFA



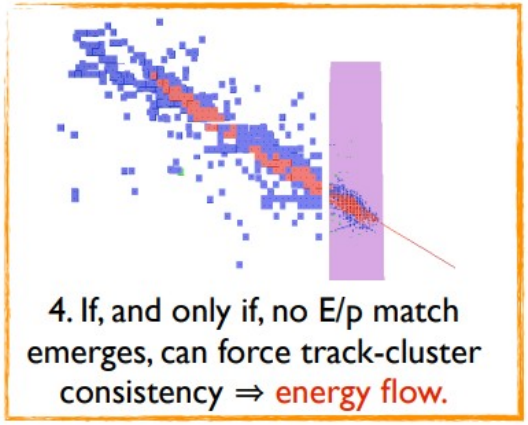
1. Multiple tracks associated to single cluster – split cluster.



2. Cluster energy much greater than track momentum – split cluster.



3. Track momentum much greater than cluster energy – bring in nearby clusters and reconfigure.



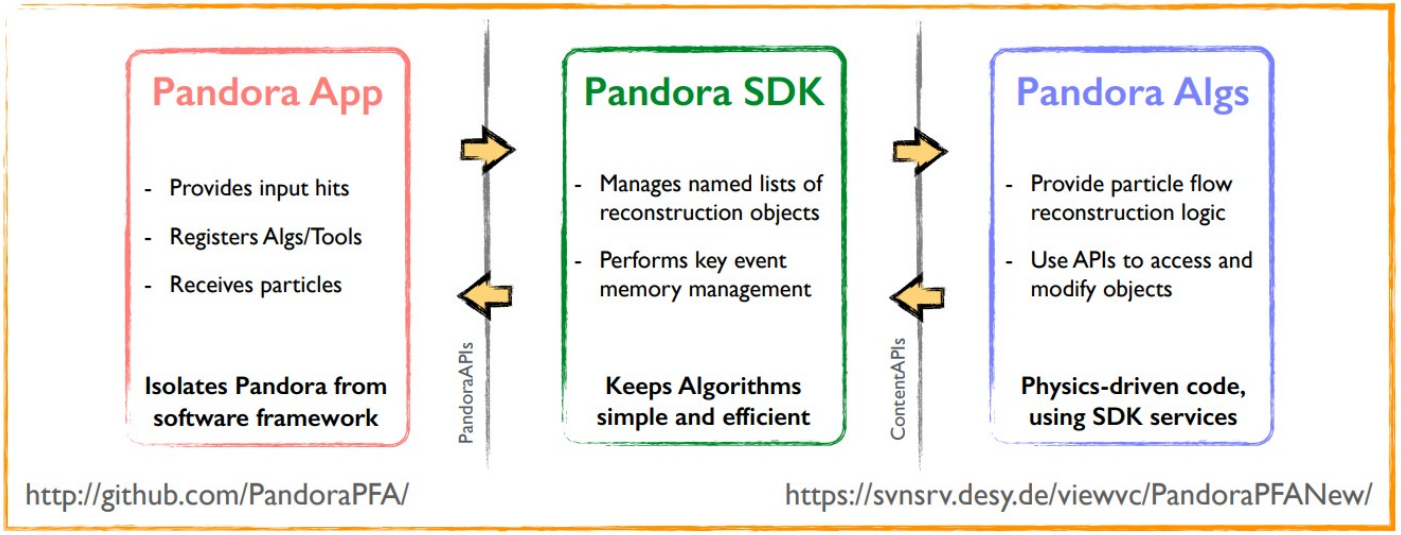
4. If, and only if, no E/p match emerges, can force track-cluster consistency ⇒ energy flow.

Pandora PFA

The Pandora Software Development Kit provides a software environment in which:

1. It is easy for users to provide the building-blocks that define a pattern recognition problem.
2. Logic required to solve pattern recognition problems is cleanly implemented in algorithms.
3. Operations to access or modify building-blocks, or to create new structures, are requested by algorithms and performed by the Pandora framework.

This promotes the use of large numbers of algorithms, each addressing specific event topologies.



http://github.com/PandoraPFA/ https://svnsrv.desy.de/viewvc/PandoraPFANew/